

Quantum Computing Cram Sheet

- **Qubit:** Can exist in superposition of $|0\rangle$ and $|1\rangle$.
- **Hadamard (H):** Creates superposition.
- **X Gate:** Bit flip.
- **Z Gate:** Phase flip.
- **CNOT:** Entangles qubits.
- **Measurement:** Collapses quantum state to classical bits.
- **Grover Workflow:** Initialize $\rightarrow$ Oracle $\rightarrow$ Diffusion $\rightarrow$ Measure.
- **Oracle:** Marks the solution by phase flipping it.
- **Diffusion:** Amplifies the marked state's probability.
- **3 qubits:** 8 basis states (2^3).
- **Common mistake:** Measuring before the oracle/diffusion.
- **Hardware:** IBM/Google/Rigetti = superconducting; IonQ/Quantinuum = trapped ion; D-Wave = quantum annealing.